

TUC is:[2D50-2E2Z](#)

Malignant
neoplasm
metastases

code to:

Primary site of malignant neoplasm if known. If the primary site of malignant neoplasm is not known or indicated, code to malignant neoplasm without specification of site [2D40-2D4Z](#) *Malignant neoplasms of ill-defined or unspecified primary sites.*

TUC is:

[5A20.0](#) - [5A2Y](#) Acute complications
of diabetes mellitus

code to:

[5A10-5A13](#) if type of diabetes is stated **or code to**
[5A14](#) Diabetes mellitus, type unspecified, if the type
of diabetes is not stated

[8B92.2](#) Diabetic lumbosacral
plexopathy

[8B94](#) Diabetic
radiculoplexoneuropathy

[8C03.0](#) Diabetic polyneuropathy

[8D88.1](#) Autonomic neuropathy due
to diabetes mellitus

[9B10.21](#) Diabetic cataract

[9B71.00](#) - [9B71.0Z](#) Diabetic
retinopathy

[BC43.7](#) Diabetic cardiomyopathy

[BD54](#) Diabetic foot ulcer

[EB90.0](#) Diabetic skin lesions

[FA38.0](#) Diabetic arthropathy

[FA38.10](#) Diabetic Charcot
arthropathy

[MF83](#) Diabetic glomerular changes

TUC is:

[5D40-5D46](#) Postprocedural
endocrine or metabolic disorders

[8E60-8E67](#) Postprocedural
disorders of the nervous system

[9D20-9D25](#) Postprocedural
disorders of eye or ocular adnexa

[AB90-AB93](#) Postprocedural
disorders of ear or mastoid process

[BE10-BE1F.1](#) Postprocedural
disorders of circulatory system

[CB60-CB64](#) Postprocedural
disorders of the respiratory system

[DE10-DE13](#) Postprocedural
disorders of digestive system

[FC01](#) Postprocedural disorders of
the musculoskeletal system

[GC70-GC7C](#) Postprocedural
disorders of genitourinary system

See Section [2.18.4 Step M4 - Instructions on medical procedures, main injury, poisoning, and maternal deaths](#)

TUC is:

[6C40.3](#) Alcohol
intoxication

code to:

[PB30](#) Unintentional exposure to or harmful effects of alcohols if
Manner of death stated on MCCD is disease

[PB30](#) Unintentional exposure to or harmful effects of alcohols if
Manner of death stated on MCCD is accident

[PD00](#) Intentional self-harm by exposure to or harmful effects of
alcohols if Manner of death stated on MCCD is intentional self-harm

[PE90](#) Assault by exposure to or harmful effects of alcohols if Manner
of death stated on MCCD is assault

[PH50](#) Exposure to or harmful effects of undetermined intent of
alcohols if Manner of death stated on MCCD could not be
determined

[PB30](#) Unintentional exposure to or harmful effects of alcohols if
Manner of death stated on MCCD is pending investigation

[PB30](#) Unintentional exposure to or harmful effects of alcohols if
Manner of death stated on MCCD is unknown

TUC is:

[6C43.3](#) Opioid
intoxication

code to:

[PB20](#) Unintentional exposure to or harmful effects of opioids or
related analgesics if Manner of death stated on MCCD is disease

[PB20](#) Unintentional exposure to or harmful effects of opioids or
related analgesics if Manner of death stated on MCCD is accident

[PC90](#) Intentional self-harm by exposure to or harmful effects of
opioids or related analgesics if Manner of death stated on MCCD is
intentional self-harm

[PE80](#) Assault by exposure to or harmful effects of opioids or related
analgesics if Manner of death stated on MCCD is assault

[PH40](#) Exposure to or harmful effects of undetermined intent of
opioids or related analgesics if Manner of death stated on MCCD
could not be determined

[PB20](#) Unintentional exposure to or harmful effects of opioids or
related analgesics if Manner of death stated on MCCD is pending
investigation

[PB20](#) Unintentional exposure to or harmful effects of opioids or
related analgesics if Manner of death stated on MCCD is unknown

TUC is:

[6C41.3](#) Cannabis intoxication [6C42.3](#) Synthetic cannabinoid intoxication
[6C49.3](#) Hallucinogen intoxication

code to:

[PB23](#) Unintentional exposure to or harmful effects of cannabinoids or hallucinogens if Manner of death stated on MCCD is disease

[PB23](#) Unintentional exposure to or harmful effects of cannabinoids or hallucinogens if Manner of death stated on MCCD is accident

[PC93](#) Intentional self-harm by exposure to or harmful effects of cannabinoids or hallucinogens if Manner of death stated on MCCD is intentional self-harm

[PE83](#) Assault by exposure to or harmful effects of cannabinoids or hallucinogens if Manner of death stated on MCCD is assault

[PH43](#) Exposure to or harmful effects of undetermined intent of cannabinoids or hallucinogens if Manner of death stated on MCCD could not be determined

[PB23](#) Unintentional exposure to or harmful effects of cannabinoids or hallucinogens if Manner of death stated on MCCD is pending investigation

[PB23](#) Unintentional exposure to or harmful effects of cannabinoids or hallucinogens if Manner of death stated on MCCD is unknown

TUC is:

[6C45.3](#) Cocaine intoxication [6C46.3](#) Stimulant intoxication including amphetamines, methamphetamine or methcathinone [6C47.3](#) Synthetic cathinone intoxication [6C48.2](#) Caffeine intoxication [6C4C.3](#) MDMA or related drug intoxication, including MDA

code to:

[PB22](#) Unintentional exposure to or harmful effects of psychostimulants if Manner of death stated on MCCD is disease

[PB22](#) Unintentional exposure to or harmful effects of psychostimulants if Manner of death stated on MCCD is accident

[PC92](#) Intentional self-harm by exposure to or harmful effects of psychostimulants if Manner of death stated on MCCD is intentional self-harm

[PE82](#) Assault by exposure to or harmful effects of psychostimulants if Manner of death stated on MCCD is assault

[PH42](#) Exposure to or harmful effects of undetermined intent of psychostimulants if Manner of death stated on MCCD could not be determined

[PB22](#) Unintentional exposure to or harmful effects of psychostimulants if Manner of death stated on MCCD is pending investigation

[PB22](#) Unintentional exposure to or harmful effects of psychostimulants if Manner of death stated on MCCD is unknown

TUC is:

[6C44.3](#) Sedative, hypnotic or anxiolytic intoxication [6C4B.3](#) Volatile inhalant intoxication [6C4D.3](#) Dissociative drug intoxication including Ketamine or PCP

code to:

[PB21](#) Unintentional exposure to or harmful effects of sedative hypnotic drugs or other CNS depressants if Manner of death stated on MCCD is disease

[PB21](#) Unintentional exposure to or harmful effects of sedative hypnotic drugs or other CNS depressants if Manner of death stated on MCCD is accident

[PC91](#) Intentional self-harm by exposure to or harmful effects of sedative hypnotic drugs or other CNS depressants if Manner of death stated on MCCD is intentional self-harm

[PE81](#) Assault by exposure to or harmful effects of sedative, hypnotic drugs or other CNS depressants if Manner of death stated on MCCD is assault

[PH41](#) Exposure to or harmful effects of undetermined intent of sedative hypnotic drugs or other CNS depressants if Manner of death stated on MCCD could not be determined

[PB21](#) Unintentional exposure to or harmful effects of sedative hypnotic drugs or other CNS depressants if Manner of death stated on MCCD is pending investigation

[PB21](#) Unintentional exposure to or harmful effects of sedative hypnotic drugs or other CNS depressants if Manner of death stated on MCCD is unknown

TUC is:

[6C4E.3](#) Other specified psychoactive substance intoxication [6C4G.3](#)

Intoxication due to unknown or unspecified psychoactive substance

code to:

[PB28](#) Unintentional exposure to or harmful effects of other or unspecified drug, medicament or biological substance if Manner of death stated on MCCD is disease

[PB28](#) Unintentional exposure to or harmful effects of other or unspecified drug, medicament or biological substance if Manner of death stated on MCCD is accident

[PC98](#) Intentional self-harm by exposure to other and unspecified drug, medicament and biological substance if Manner of death stated on MCCD is intentional self-harm

[PE88](#) Assault by exposure to or harmful effects of other or unspecified drug, medicament or biological substance if Manner of death stated on MCCD is assault

[PH48](#) Exposure to or harmful effects of undetermined intent of other or unspecified drugs, medicaments or biological substances if Manner of death stated on MCCD could not be determined

[PB28](#) Unintentional exposure to or harmful effects of other or unspecified drug, medicament or biological substance if Manner of death stated on MCCD is pending investigation

[PB28](#) Unintentional exposure to or harmful effects of other or unspecified drug, medicament or biological substance if Manner of death stated on MCCD is unknown

TUC is:	code to:
6D80 Dementia due to Alzheimer disease	8A20 , Alzheimer disease
6D81 Dementia due to cerebrovascular disease	The originating cerebrovascular disease in Chapter 08 ' 8B00 to 8B2Z Cerebrovascular diseases, unspecified' and if not reported code to 8B2Z Cerebrovascular diseases, unspecified
6D82 Dementia due to Lewy body disease	8A22 , Lewy body disease
6D83 Frontotemporal dementia	8A23 , Frontotemporal lobar degeneration
6D84.0 Dementia due to use of alcohol	The originating disorder due to use of alcohol, if not reported code to 6C40.Y , other specified disorders due to use of alcohol
6D84.1 Dementia due to use of sedatives, hypnotics or anxiolytics	The originating disorder due to use of sedative, hypnotic or anxiolytic, if not reported code to 6C44.Y , other specified disorders due to use of sedative, hypnotic or anxiolytic
6D84.2 Dementia due to use of volatile inhalants	The originating disorder due to use of volatile inhalants, if not reported code to 6C4B.Y , other specified disorders due to use of volatile inhalant
6D84.Y Dementia due to other specified psychoactive substance	The originating disorder due to other specified psychoactive substance, if not reported code to 6C4G.Y Other specified disorders due to use of unknown or unspecified psychoactive substances
6D85.0 Dementia due to Parkinson disease	8A00.0Z , Parkinson disease, unspecified
6D85.1 Dementia due to Huntington disease	8A01.10 Huntington disease, unspecified
6D85.2 Dementia due to exposure to heavy metals and other toxins	PB36 Unintentional exposure to or harmful effects of other or unspecified substances chiefly nonmedicinal as to source PB80-PD3Z Intentional self-harm, unspecified PE95 Assault exposure to or harmful effects of other or unspecified substances chiefly nonmedicinal as to source PH56 Exposure to or harmful effects of other or unspecified substances chiefly nonmedicinal as to source, undetermined intent
6D85.3 Dementia due to human immunodeficiency virus	1C62.3 , HIV disease clinical stage 4 without mention of tuberculosis or malaria
6D85.4 Dementia due to multiple sclerosis	8A40.Z , Multiple sclerosis, unspecified
6D85.5 Dementia due to prion disease	Prion disease mentioned, 8E0Z Human prion diseases, unspecified

TUC is:	code to:
6D85.6 Dementia due to normal pressure hydrocephalus	8D64.04 Normal-pressure hydrocephalus
6D85.7 Dementia due to injury to the head	External cause that caused head injury, if unspecified Code to: PB6Z Unspecified unintentional cause of morbidity or mortality PD05 Intentional self-harm , unspecified PF2Z Assault, unspecified PH8Y Other specified injury event of undetermined intent
6D85.8 Dementia due to pellagra	5B5C.0 , Pellagra
6D85.9 Dementia due to Down syndrome	LD40.0 , Complete trisomy 21
6D85.Y Dementia due to other specified diseases classified elsewhere	specified disease
6D8Y Dementia, other specified cause	specified cause, if unable to code the specified cause, code to 6D8Z Dementia, unknown or unspecified cause

TUC is:	code to:
BA42 Subsequent myocardial infarction	BA41.Z Acute myocardial infarction, unspecified
(BA60 ,-) Certain current complications following acute myocardial infarction	BA41.Z Acute myocardial infarction, unspecified

TUC is:	code to:
BA43 Coronary thrombosis not resulting in myocardial infarction	BA41.Z Acute myocardial infarction, unspecified. For mortality, the occurrence of myocardial infarction is assumed.

TUC is:	code to:
BA50 Old myocardial infarction	BA5Z Chronic ischaemic heart disease, unspecified, if the cause is not stated

TUC is:	code to:
BD55 Asymptomatic stenosis of intracranial or extracranial artery	(8B11 ,-)Cerebral ischaemic stroke
BD56 Asymptomatic occlusion of intracranial or extracranial artery	(8B11 ,-) Cerebral ischaemic stroke

TUC is: JA05 Complications following abortion, ectopic or molar pregnancy	code to: JA00-JA0Z Abortive outcome of pregnancy except JA05 Complications following abortion, ectopic or molar pregnancy
TUC is: JB65 Sequelae of complication of pregnancy, childbirth or the puerperium	code to: JB62.Z Death from sequelae of obstetric causes, unspecified
TUC is: KB60-KB6Z Transitory endocrine or metabolic disorders specific to Fetus or newborn, except KB60.0 Syndrome of infant of mother with gestational diabetes; KB60.1 Syndrome of infant of a diabetic mother, type 1 or 2, nongestational, insulin dependent; KB60.2 Neonatal diabetes mellitus; or KB62.0 Transitory neonatal hyperthyroidism	code to: other perinatal cause. If no other perinatal cause is reported, KD5Z Conditions originating in the perinatal or neonatal period, unspecified
TUC is: MA13.1 Finding of alcohol in blood	code to: MH14 Other ill-defined and unspecified causes of mortality
TUC is: MA15.Y Other specified microbiological findings in blood, blood-forming organs, or the immune system	code to: The originating infectious disease in Chapter 01 'Certain infectious or parasitic diseases', or to 1G40 Sepsis without septic shock - 1G41 Sepsis with septic shock.
TUC is: MG20.0 Malignant cachexia	code to: 2D4Z Unspecified malignant neoplasms of unspecified sites
TUC is: MG48 Unknown and unspecified causes of morbidity	code to: MH10 - MH14 Ill-defined and unknown causes of mortality

TUC is:

[MG50-MG5Z](#) Finding of microorganism resistant to antimicrobial drugs

code to:

The infection of unspecified site for the microorganism. If the microorganism is not specified, **code to** [1H0Z](#) *Infection, unspecified*

TUC is:

Chapter 22 Injury, poisoning or certain other consequences of external causes

Not to be used for the underlying cause of death, except as an additional code to the relevant category in Chapter 23 (see also Step M4).

If there is no corresponding code in Chapter 23, see guidance at 2.21.8.3 for additional information to determine the correct code from chapter 23.

TUC is:

24 Factors influencing health status or contact with health services

Not to be used during the selection of the underlying cause of death

Code to: [MH14](#) Other ill-defined and unspecified causes of mortality **if nothing else is reported on the certificate**

TUC is:

Chapter X Extension codes, **except** infectious agents

code to:

[MH14](#) Other ill-defined and unspecified causes of mortality

TUC is:

Infectious agents in Chapter X Extension codes

code to:

the infection of the infectious agent of unspecified site. See also [2.21.8.11](#).

2.19.3.25 Codes not to be used if the underlying cause is known or other specific conditions apply

For the conditions specified in the left-hand column of this section, if the condition specified in the right-hand column **does not apply**, then keep the condition as the tentative underlying cause.

TUC is:

[2D43](#) Malignant neoplasms of independent, multiple primary sites

Not to be used for the underlying cause of death, if the multiple neoplasms are reported separately.

Code to: [2A00-2A0Z](#) Neoplasms of brain or central nervous system; [2A20-2B3Z](#) Neoplasms of haematopoietic or lymphoid tissues; or [2B50-2D3Z](#) Malignant neoplasms, stated or presumed to be primary, of specified sites, except of lymphoid, haematopoietic, central nervous system or related tissues

TUC is:

[6A00](#) Disorders of intellectual development [6A01](#) Developmental speech or language disorders [6A03](#) Developmental learning disorder [6E60-6E6Z](#) Secondary mental or behavioural syndromes associated with disorders or diseases classified elsewhere

Not to be used for the underlying cause of death, if the underlying physical condition is known.

Code to: the underlying physical condition.

TUC is:

[6C4A](#) Disorders due to use of nicotine

Not to be used for the underlying cause of death, if the resultant physical condition is known.

Code to: the resultant physical condition.

TUC is:

[BC00](#) Multiple valve disease

Not to be used for underlying cause of death, if the multiple valve diseases are reported separately and nonrheumatic origin.

Code to: each valve disease specified by the coding tool, and select the underlying cause by applying the selection and modification rules in the normal way.

TUC is:

[JA80](#) Maternal care related to multiple gestation

Not to be used for the underlying cause of death, if a more specific complication is reported.

Code to: the more specific complication.

If there is no specific complication, **code to** [JB0Z](#) Complications of labor or delivery, unspecified

TUC is:

[KA21](#) Disorders of newborn related to short gestation or low birth weight, not elsewhere classified
[KA22](#) Disorders of newborn related to long gestation or high birth weight

Not to be used for the underlying cause of death, if any other cause of perinatal mortality or any developmental anomaly is reported. This does not apply for [KB2D](#) Respiratory failure of newborn or [KB2E](#) Respiratory Arrest of newborn, which are ill-defined conditions and should not be used for underlying cause of death.

Code to: the other specified cause of perinatal mortality, or developmental anomaly
([19](https://icd.who.int/browse/latest-release/mms/en#1306203631)] Chapter 19 Certain conditions originating in the perinatal period except for [KB2D](#) Respiratory failure of newborn or [KB2E](#) Respiratory arrest of newborn

TUC is:

[KD3B](#) Fetal death, cause not specified

Not to be used for the underlying cause of death of live births.

Code to: [KD5Z](#) Conditions originating in the perinatal or neonatal period, unspecified, if the child was born alive but the cause death is unknown. See also: [2.21.8.4] (#special-instructions-on-fetal-death)

TUC is:

[KD3B](#) Fetal death, cause not specified

Not to be used for the underlying cause of fetal death, if any other cause of fetal mortality is known.

Code to: the other cause of fetal mortality.

TUC is:

[MB50-MB5Z](#) Paralytic symptoms

Not to be used for the underlying cause of death, if the cause of the paralysis is known.

Code to: the cause of the paralysis.

2.19.4 Special instructions on surgery and other medical procedures (Step M4)

2.19.4.1 Reason for the surgery or procedure stated

If the tentative underlying cause selected by applying Steps SP1 to SP8 and M1 to M3 is surgery or other medical procedure and the certificate states the reason for which the operation or procedure was performed, then select the reason for the operation or

procedure as the new tentative underlying cause of death. Next, reapply the instructions in Steps SP7 and M1 to M4.

2.19.4.2 Reason for the surgery or procedure not stated, complication reported

If the reason for the surgery or procedure is not stated and a complication is reported, proceed as described next.

- a) Surgery indicates specific organ: First, if the type of surgery or procedure indicates a specific organ or site, then use the code for the residual category for the organ or site operated on as the new tentative underlying cause of death. Next, reapply the instructions in Steps SP7 and M1 to M4.
- b) If above does not apply, then use the appropriate code from:
 - [JB0C](#) *Complications of anaesthesia during labour or delivery*
 - [JB0D.3](#) *Other complications of obstetric surgery or procedures or*
 - [PK80-PK8Z](#) *Surgical or other medical procedures associated with injury or harm in diagnostic or therapeutic use*
 - [PL11](#) *Mode of injury or harm associated with a surgical or other medical procedure*

When both [PK80-PK8Z](#) and [PL11](#) applies, then code the mode of injury or harm ([PL11](#)) first and add the type of surgery or procedure [PK80-PK8Z](#) to the cluster.

2.19.4.3 Reason for the surgery or procedure not stated, no complication reported

If the reason for the surgery or procedure is not stated and no complication is reported, proceed as described next:

- a) Surgery indicates specific organ: If the type of surgery or procedure indicates a specific organ or site, then use the code for the residual category for the organ or site operated on as the new starting point. Next, reapply the instructions in Steps SP7 and M1 to M4.
- b) Lastly, if above does not apply, code to [MH14](#) *Other ill-defined or unspecified causes of mortality*.

Example 1

- 1 (a) Postoperative haemorrhage
 - (b) Caesarean section
 - (c)
 - (d)
- 2

Reason for surgery Prolonged labour

The certificate states the reason why the surgery was performed. Code the reason for the surgery, prolonged labour, as the underlying cause of death [JB03.Z](#) *Long labour, unspecified*.

Example 2

- 1 (a) Pulmonary embolism
 - (b) Appendectomy
 - (c)
 - (d)
- 2

The certificate does not specify the reason for the surgery, but a complication of the surgery pulmonary embolism is reported. The term appendectomy indicates appendix as the organ operated on. Code [DB1Z Diseases of appendix, unspecified](#) as the underlying cause of death.

Example 3

- 1 (a) Unintentional puncture of aorta
 - (b) Laparotomy
 - (c)
 - (d)
- 2

The certificate does not specify the reason for the surgery and the term laparotomy does not indicate a specific organ. However, there is a mention of a mode of injury at the time of the surgery. Code the mode of injury, unintentional puncture during laparotomy as the underlying cause of death [PL11.0 Cut, puncture or tear, as mode of injury or harm](#).

Example 4

- 1 (a) Laparotomy
 - (b)
 - (c)
 - (d)
- 2

The certificate does not specify why the surgery was performed and the term laparotomy does not indicate a specific organ. There is no mention of a complication. Code [MH14 Other ill-defined or unspecified causes of mortality](#), as the underlying cause of death.

2.19.4.4 Medical devices associated with adverse incidents due to external causes

If a death is caused by an incident involving a medical device, but the incident is due to an external cause and not to any breakdown or malfunctioning of the device itself, code the external cause as the underlying cause of death.

If the external cause of the incident is not specifically classified, code to [PB6Z Unspecified unintentional cause of morbidity or mortality](#) (See example 3).

Example 1

- 1 (a) Inhalation pneumonia
(b) Haemorrhage of trachea
(c) Fell from bed while attached to respirator
(d)
- 2 Respiratory treatment following liver transplant

There is no mention of breakdown or malfunctioning of the respirator or the tracheal tube. Code [PL14.E](#) *Fall in health care*, the accident that caused the haemorrhage, as the underlying cause of death, and additional code, if desired, for [XE8PK](#) *Bed, bedding or bedding accessories*.

Example 2

- 1 (a) Pulmonary oedema
(b) Intra-aortic balloon pump stopped
(c) Power cut due to hurricane
(d) Recent myocardial infarction with mitral insufficiency
- 2

The balloon pump stopped working, not because of any malfunctioning or breakdown, but because of a power cut. Code the reason of the power cut, cataclysmic storm, as the underlying cause of death, ([PJ06](#)).

Example 3

- 1 (a) Cardiac and respiratory failure
(b) Stopped administration of inotropic drugs
(c) Accidental removal of subclavian line
(d)
- 2 Surgery for acute rupture of gallbladder

There is no mention of malfunctioning or breakdown of equipment. Since the accident that caused the removal of the subclavian line is not described, code to [PB6Z](#) *Unspecified unintentional cause of morbidity or mortality*.

2.19.5 Special instructions on main injury in deaths from external causes (Step M4)

If the underlying cause selected by applying the selection and modification rules in Steps SP1 to SP8 and M1 to M3 is an injury, code the external cause of the injury as the underlying cause of death.

In addition to the underlying cause from Chapter 23 'External causes of morbidity and mortality', also code a main injury. If an injury is recorded but not an external cause, first infer the intent (sometimes called manner of death), see Section [2.21.8.8 Intent of external causes](#) then code the external cause using Table . Table provides default codes to be used only if there is insufficient information to select a more specific code in Chapter 23, taking into account the contents of the MCCD and any other sources of information that may be available according to national practice.

This applies to both body injuries and poisoning. For special instructions on how to identify the underlying cause and main injury in poisoning deaths, see Section [2.19.6 Special instructions on poisoning by drugs, medications and biological substances \(Step M4\)](#).

If more than one injury is reported on the death certificate, apply the following instructions:

- (a) When the injuries reported include trivial injuries (those listed in Annex [3.14.10 List of conditions unlikely to cause death](#)), whether in Part 1 or Part 2, select the main injury as if the injuries in the list of Annex [3.14.10](#) had not been reported.

Example 1

- 1
 - (a) Contusion of arm and fracture of skull
 - (b) Fall from scaffolding
 - (c)
 - (d)
- 2

Fall from scaffolding is the underlying cause of death. Code underlying cause to [PA61 Unintentional fall from a height of 1 metre or more](#) and use additional code, if desired, for the [XE7RK Scaffolding](#). As main injury, code [NA02.Z Fracture of skull or facial bones, part unspecified](#). Disregard contusion of arm (Superficial injury of upper limb, level unspecified), as it is in the Annex [3.14.10 List of conditions unlikely to cause death](#).

- (b) When non-trivial injuries are reported in both Part 1 and Part 2, select the main injury from Part 1. This applies even when the injuries mentioned in Part 2 have a higher rank in Annex [3.14.5 Priority ranking of Nature-of-Injury codes](#), than the injuries mentioned in Part 1.

Example 2

- 1
 - (a) Multiple intrathoracic injuries
 - (b) Car driver, collision with bus
 - (c)
 - (d)
- 2 Brain injuries

Code to [PA04 Unintentional land transport traffic event injuring a car occupant](#), and use additional code, if desired, for [XE5LJ Bus or coach as counterpart in land transport crash](#). As main injury, code [NB32.7 Multiple injuries of intrathoracic organs](#). Unspecified brain injury has a higher rank in Annex [3.14.5 Priority ranking of Nature-of-Injury codes](#) than multiple injuries of thorax, but multiple injuries of thorax are mentioned in Part 1 and take precedence over the injuries mentioned in Part 2.

- (c) When non-trivial injuries are reported only in Part 2, select a main injury from Part 2.
- (d) When more than one serious injury is reported in the relevant part of the certificate, select the main injury according to Annex [3.14.5 Priority ranking of Nature-of-Injury codes](#). Note that 1 is the highest priority rank and that 6 the lowest.

Example 3

- 1 (a) Multiple intrathoracic injuries and brain injuries
- (b) Car driver, collision with bus
- (c)
- (d)

2

Code to [PA04](#) *Unintentional land transport traffic event injuring a car occupant* as underlying cause of death. As main injury, code brain injury [NA07.Z](#) *Intracranial injury, unspecified*, which has a higher rank on the priority list than [NB32.7](#) *Multiple injuries of intrathoracic organs*.

- (e) When more than one of the serious injuries reported in the relevant part of the certificate have the same and highest rank, select the first mentioned of these injuries. However, select a specific injury over an injury from the group [ND30-ND37](#) *Injuries involving multiple body regions* with the same priority rank.

Example 4

- 1 (a) Multiple injuries with traumatic subdural haemorrhage
- (b) Car driver, collision with bus
- (c)
- (d)

2

Code to [PA04](#) *Unintentional land transport traffic event injuring a car occupant* as underlying cause of death. As main injury, code [NA07.6Z](#) *Traumatic subdural haemorrhage, unspecified whether acute or chronic*. Multiple injuries and traumatic subdural haemorrhage have the same rank on the priority list, but a specific injury takes precedence over injury from the group Injuries involving multiple body regions.

2.19.6 Special instructions on poisoning by drugs, medications and biological substances (Step M4)

If poisoning is the tentative underlying cause in Step M4 and multiple substances are reported, follow the instructions in this section.

2.19.6.1 The drug most likely to have caused death is specified

If one of the substances is specified as the substance most likely to have caused the death, code the external cause code for that substance as the underlying cause of death. Use additional code from Chapter X, if applicable, to identify the specific substance reported, and add the main injury from Chapter 22 to the cluster.

Example 1

- 1 (a) Unintentional heroin overdose
 - (b)
 - (c)
 - (d)
- 2 Diazepam and amitriptyline present

By placing heroin overdose alone in Part 1 and reporting the other substances as contributing causes of death in Part 2, the certifier has identified heroin as the substance most likely to have caused the death.

Select [PB20](#) *Unintentional exposure to or harmful effects of opioids or related analgesics* as underlying cause. Use additional code [XM05B3](#) *Diamorphine* to identify the specific substance reported. Add main injury from Chapter 22 [NE60](#) *Harmful effects of drugs, medicaments or biological substances, not elsewhere classified*, NEC to the cluster. The cluster is [PB20&XM05B3/NE60](#).

Example 2

- 1 (a) Poisoning by amphetamine
(b)
(c)
(d)
- 2 Toxic levels of heroin and flunitrazepam

By placing amphetamine poisoning alone in Part 1 and reporting the other substances as contributing causes of death in Part 2, the certifier has identified amphetamine as the substance most likely to have caused the death. Select [PB22](#) *Unintentional exposure to or harmful effects of psychostimulants* as underlying cause. Use additional code [XM48Z9](#) *Amphetamine* to identify the specific substance reported. And add [NE60](#) *Harmful effects of drugs, medicaments or biological substances, not elsewhere classified* to the cluster. The cluster is [PB22&XM48Z9/NE60](#).

Example 3

- 1 (a) Poisoning by alcohol
(b)
(c)
(d)
- 2 Toxic levels of heroin and flunitrazepam

By placing alcohol poisoning alone in Part 1 and reporting the other substances as contributing causes of death in Part 2, the certifier has identified alcohol as the substance most likely to have caused the death. Select [PB30](#) *Unintentional exposure to or harmful effects of alcohols* as underlying cause. And add [NE61](#) *Harmful effects of or exposure to noxious substances, chiefly nonmedicinal as to source, not elsewhere classified* to the cluster. The cluster is [PB30/NE61](#).

Example 4

- 1 (a) Alcohol poisoning
(b)
(c)
(d)
- 2 Diazepam and amitriptyline present

By placing alcohol poisoning alone in Part 1 and reporting the other substances as contributing causes of death in Part 2, the certifier has identified alcohol as the most important substance in bringing about the death. Select [PB30](#) *Unintentional exposure to or harmful effects of alcohols* as underlying cause. And add [NE61](#) *Harmful effects of or exposure to noxious substances, chiefly nonmedicinal as to source, not elsewhere classified* to the cluster. The cluster is [PB30/NE61](#).

2.19.6.2 The drug most likely to have caused death is not specified

If multiple substances are reported as contributing to the death but none of the substance is specified as the substance most likely to have caused the death, follow these instructions:

- (a) Code combinations of alcohol with a drug to the drug

Example 5

- 1 (a) Toxic levels of alcohol and flunitrazepam
(b)
(c)
(d)
- 2 Diazepam and amitriptyline present

By placing toxic levels of alcohol and flunitrazepam in Part 1 and reporting the other substances as contributing causes of death in Part 2, the certifier has identified alcohol and flunitrazepam as the most important substances in bringing about the death. Of these two, select poisoning by flunitrazepam because combinations of alcohol with a drug are coded to the drug. Select [PB21](#) *Unintentional exposure to or harmful effects of sedative hypnotic drugs or other CNS depressants* as underlying cause. Use additional code [XM9W71](#) *Flunitrazepam* to identify the specific substance reported. And add [NE60](#) *Harmful effects of drugs, medicaments or biological substances, not elsewhere classified*, NEC. The cluster is [PB21&XM9W71/NE60](#).

- (b) Code combinations of multiple drugs, as follows:

- If the external cause of the multiple drugs reported is the same select that as the underlying cause of death.
- If the external cause of the multiple drugs reported is not the same, code [PB29](#) *Unintentional exposure to or harmful effects of multiple drugs, medicaments or biological substances* as the underlying cause of death.

Use additional code from Chapter X, if applicable, to identify the substance most likely to have caused the death by referring to Section [2.19.6.3 Identification of the drug most likely to have caused death](#).

Note that when adding more than one drugs in optional use cases, the substance most likely to have caused the death identified as above must be coded first.

Example 6

- 1 (a) Toxic levels of heroin and amphetamine
(b)
(c)
(d)
- 2

Neither heroin nor amphetamine are identified as the substance most likely to have caused the death and the external cause of these drugs are not the same. Code to [PB29](#) *Unintentional exposure to or harmful effects of multiple drugs, medicaments or biological substances* as the underlying cause of death. Go to Section [2.19.6.3](#) to identify the drug most likely to have caused the death.

Example 7

- 1 (a) Unintentional poisoning by alcohol, heroin, and diazepam
- (b)
- (c)
- (d)

2

None of the substances is identified as the substance most likely to have caused the death. Poisoning by combinations of alcohol and drugs are coded to the drugs. Because none of the drugs is identified as most important, and the external cause code is different, code to [PB29](#) *Unintentional exposure to or harmful effects of multiple drugs, medicaments or biological substances* as the underlying cause of death. And go to Section [2.19.6.3](#) to identify the substance most likely to have caused the death.

2.19.6.3 Identification of the drug most likely to have caused death

Use the priority order below to identify the substance most likely to have caused the death (1 = highest priority):

1. Opioid agonists and partial agonists and other and unspecified narcotics. Deaths that include multiple opioids classifiable should be prioritised as:
 - 1a. Heroin
 - 1b. Methadone
 - 1c. Opium
 - 1d. Other opioids
 - 1e. Other synthetic narcotics
 - 1f. Other and unspecified narcotics
2. Inhaled and intravenous anaesthetic agents, Includes: Propofol
3. Tricyclic and tetracyclic antidepressants
4. Barbiturates
5. 4-Aminophenolderivatives Includes: APAP, acetaminophen, paracetamol
6. Antipsychotics and neuroleptics Includes: Phenothiazine antipsychotics and neuroleptics, Butyrophenone and thioxanthene neuroleptics, Other and unspecified antipsychotics and neuroleptics
7. Antiepileptic drugs, antiparkinsonism drugs and unspecified sedatives
8. Cocaine
9. Psychostimulants with abuse potential Includes: Amphetamines and derivatives
10. Monoamine oxidase inhibitor (MAO) antidepressants and other and unspecified antidepressants. Includes: Selective serotonin reuptake inhibitors (SSRIs), venlafaxine
11. Benzodiazepines
12. Drugs and substances not listed above

If there is more than one drug in the same priority group, code to the first mentioned.

Example 8

- 1 (a) Toxic levels of cocaine, heroin, diazepam, and amitriptyline
- (b)
- (c)
- (d)

2

None of the drugs is identified as the substance most likely to have caused the death, and the external cause code is not the same for these substances. Code to [PB29](#) *Unintentional exposure to or harmful effects of multiple drugs, medicaments or biological substances* as the underlying cause of death. On the priority list above, cocaine is in group 8, heroin is in group 1a, diazepam is in group 11 and amitriptyline is in group 3. Use additional code [XM05B3](#) *Diamorphine* for the drug identified ([PB29](#) *Unintentional exposure to or harmful effects of multiple drugs, medicaments or biological substances* & [XM05B3](#)). Add codes, if desired, from Chapter X to list other drugs reported. Finally, add [NE60](#) *Harmful effects of drugs, medicaments or biological substances, not elsewhere classified*, to the cluster ([PB29](#) & [XM05B3](#) / [NE60](#)).

Example 9

- 1 (a) Heroin, cocaine, diazepam and amitriptyline overdose
- (b)
- (c)
- (d)

2

None of the drugs is identified as the substance most likely to have caused the death, and the external cause code is not the same for these substances. Code to [PB29](#) *Unintentional exposure to or harmful effects of multiple drugs, medicaments or biological substances* as the underlying cause of death. On the priority list above, heroin is in group 1a, cocaine is in group 8, diazepam is in group 11 and amitriptyline is in group 3. Use additional code [XM05B3](#) *Diamorphine* for the drug identified. Add codes, if desired, from Chapter X to list other drugs reported. Finally, add [NE60](#) *Harmful effects of drugs, medicaments or biological substances, not elsewhere classified* to the cluster. ([PB29](#) & [XM05B3](#) / [NE60](#))

Example 10

- 1 (a) Unintentional poisoning by alcohol, heroin and diazepam
- (b)
- (c)
- (d)

2

Poisoning by combinations of alcohol and drug(s) is coded to the drug(s), see instruction in Section [2.19.6.2](#), above. None of the drugs reported in Part 1 is identified as the substance most likely to have caused the death, and the external cause code is not the same for these substances. Code to [PB29](#) *Unintentional exposure to or harmful effects of multiple drugs, medicaments or biological substances* as the underlying cause of death. On the priority list above, heroin is in group 1a and diazepam is in group 11. Use additional code [XM05B3](#) *Diamorphine* identified as most likely to have caused death. Add code [XM8P99](#) *Diazepam*, if desired. Finally, add [NE60](#) *Harmful effects of drugs, medicaments or biological substances, not elsewhere classified* to the cluster. ([PB29](#) & [XM05B3](#) & [XM8P99](#) / [NE60](#)).

2.19.7 Special instructions on maternal mortality (Step M4)

For coding of maternal mortality, follow general coding instructions.

After assigning a code for each condition reported (See Coding instructions for maternal mortality in Section [2.21.8.2](#) apply the selection and modification instructions in the normal way starting from SP1 as same as other causes of death.

Then, apply Steps SP1 to SP8 and M1 to M3 and M4 for Surgery, Injury, External causes, and Poisoning, then Step M4 for maternal mortality.

Apply the following instructions against the tentative underlying cause of death (TUC).
Terms used Terms used here in the instruction is as follows:

- Certain maternal diseases: [JA00.-](#) to [JB4Z](#), [JB60](#), [JB6Y](#), [1C14](#)
- Maternal diseases classified elsewhere: [JB63.-](#) (infections), [JB64.-](#) (other)
- Injury or external causes: Chapter 22, Chapter 23
- Other conditions: other than above

Maternal diseases

- If the tentative underlying cause (TUC) is 'certain maternal diseases' and the deceased was pregnant at the time of death, within 42 days, or the duration is unknown or unstated, keep the TUC.
- If the TUC is 'maternal diseases classified elsewhere' keep the TUC and postcoordinate the code for the specific disease classified elsewhere from Chapter 01-19.
- If the TUC is [JB61.0](#), [JB61.Z](#), [JB62.0](#) or [JB62.Z](#) keep it and its post-coordinated code, if any, as the TUC.
- If the TUC is 'certain maternal diseases' or maternal diseases classified elsewhere' but the death occurred after 42 days but less than one year after the obstetric event, then code to [JB61.-](#) as the TUC and add the code for the specific maternal disease to the cluster.

Other conditions or indirect obstetric causes

- If the TUC is [JB61.1](#) or [JB62.1](#), keep it and its post-coordinated code if any as the TUC.
- If the TUC is 'other conditions', and the pregnancy contributed to death, and the deceased was pregnant at the time of death, within 42 days, or the duration is unknown or unstated, code to [JB63.-](#) or [JB64.-](#) as appropriate and add the specific 'other condition' to the cluster.
- If the TUC is 'other conditions', and the pregnancy did not contribute to death, but the deceased was pregnant at the time of death, within 42 days, keep the TUC and add [XT0S](#) Pregnancy to the cluster.
- If the TUC is 'other conditions', and the pregnancy did not contribute to death, and the death occurred after 42 days or more, or the duration is unknown or unstated, keep the TUC.

See 'Determining whether pregnancy contributed to death' in section [2.21.8.2](#) to decide whether pregnancy contributed to death from the information on the death certificate.

Injury or external causes

- If the TUC is 'injury or external cause', and the deceased was pregnant at the time of death, within 42 days, keep the TUC and add [XT0S](#) Pregnancy to the cluster.
- If the TUC is 'injury or external cause', and the death occurred after 42 days or more, or the duration is unknown or unstated, keep the TUC.

2.20 Mortality Rules - Knowledgebase

This section explains in depth the rules guiding the choice of ICD-11 codes in mortality coding and provides special instructions about particular situations and causes of death. The rules cover, for example, permitted sequences, such as disease 'a' due to disease 'b', and cases where the selected cause may be modified to provide more relevant information for public health. Each rule is described with relevant details and examples.

Because of the large number of ICD-11 codes and code combinations which can occur in Mortality coding, WHO has published an online Mortality Rules Knowledgebase consisting of supplementary tables which give effect to the rules set out in this section of the Reference Guide. This approach reduces the need for lengthy tables of codes within the Reference Guide and allows for more convenient updating.

The Knowledgebase is intended to be accessed online manually or technically or programmatically. A reference is given in each relevant section to the corresponding table in the Knowledgebase.

2.21 Coding instructions for mortality: multiple cause coding and other specific instructions

Multiple cause coding (see also Sections [2.21.1](#) - [2.21.8](#)) permits in-depth analysis of causes of death, for example of serious but avoidable complications of certain underlying causes, and the impact of coexisting conditions on the outcome of a disease process. Therefore, in mortality coding, both underlying cause and multiple causes should be recorded. Also, complete multiple cause coding is essential for a correct application of the ICD instructions for selection and modification of the underlying cause of death (see Sections [2.17](#) - [2.19](#)).

All possible detail should be retained in the multiple cause coding, since records containing all multiple cause conditions permit more thorough analysis than records with only a selection of the conditions reported on the death certificate. In particular:

- the position of the individual codes in the data record should reflect where on the certificate the corresponding diagnostic expressions were entered by the certifier, because some analyses may focus on the terminal cause of death, or on conditions reported in Part 2
- codes for common conditions, or for conditions regarded as symptomatic or less informative, should not be deleted or left out, since they may be of special interest

in analysis of avoidable complications and may serve as markers of the seriousness of other conditions reported on the death certificate;

- multiple cause data should be stored in two formats:
 1. one format that shows as clearly as possible which term the certifier used on the certificate and where on the certificate each term was reported
 2. one format that takes the stated or implied relationships between the reported conditions into consideration, and where the codes have been harmonised according to the instructions in the ICD volumes.

Note that the syntax of a code string to retain ICD codes provided in a death certificate should be distinguishable from the syntax used for cluster coding in ICD (i.e. forward slash (/), ampersand (&)), while the specific syntax may differ according to different settings. Such code string could be for example, [BD10](#)|BA5Z*5A11/9B71.0Z, where a vertical bar (|) expresses the separator between lines in Part 1, and an asterisk expresses the separator between Part 1 and Part 2, and the forward-slash (/) shows the cluster as a separator between stems following the convention of ICD.

2.21.1 Uncertain diagnosis

Ignore expressions indicating doubt as to the certainty of the diagnosis, for example 'apparently', 'presumably', 'probably' or 'possibly'. A tentative diagnosis, although uncertain, is of better use to mortality statistics than no diagnosis at all.

2.21.1.1 Either ... or

The certifier might report alternative diagnoses, 'either diagnosis A or diagnosis B'. In such cases, proceed as follows.

2.21.1.2 One condition, either one site or another

- (a) If the sites are in the same anatomical system, code to the residual category for the group or anatomical system in which the reported sites are classified.

Example 1

- 1 (a) Cancer of kidney or bladder
 - (b)
 - (c)
 - (d)
- 2

Code as [2C9Z](#) *Malignant neoplasms of urinary tract, unspecified*.